CLIMATE

Amogy raises $80M to power ships and data centers with ammonia

Tim De Chant — 4:00 AM PDT · July 15, 2025

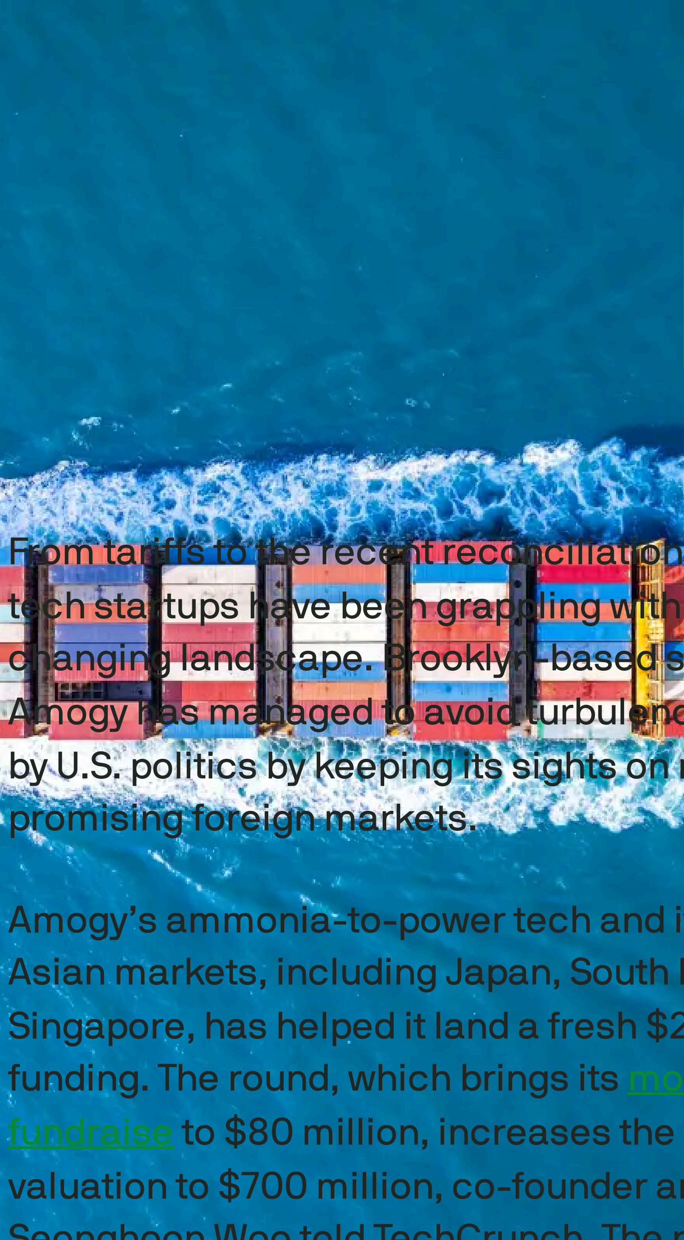

IMAGE CREDITS: GENERAL 4530 / GETTY IMAGES

From tariffs to the recent reconciliation bill, climate tech startups have been grappling with a rapidly changing landscape. Brooklyn-based startup Amogy has managed to avoid turbulence induced by U.S. politics by keeping its sights on more promising foreign markets.

Amogy's ammonia-to-power tech and its focus on Asian markets, including Japan, South Korea, and Singapore, has helped it land a fresh $23 million in funding. The round, which brings its most recent fundraise to $80 million, increases the company's valuation to $700 million, co-founder and CEO Seonghoon Woo told TechCrunch. The round was led by the Korea Development Bank and KDB Silicon Valley LLC with participation from BonAngels Venture Partners, JB Investment, and Pathway Investment.

The startup is based primarily in the U.S. but it has found demand for its core technology in Japan and South Korea, countries looking for new ways to expand power generation.

"They don't have as high-quality solar, wind, and geothermal resources, and they are not really in the best position to build a nuclear power either," Woo said.

Ammonia is most widely used as a component of plant fertilizers; it can also serve as what experts call a hydrogen carrier. Normally, hydrogen is

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difficult to transport — it's flammable and prone to leaking — but a hydrogen carrier like ammonia makes it easier.

In an effort to reduce their carbon pollution, Asian countries have started burning ammonia in existing fossil fuel power plants. Typically, operators will replace some percentage of coal with ammonia.

Shipping companies have started doing the same, replacing diesel with the compound. Ammonia has found fertile ground in that industry because the International Maritime Organization, which regulates maritime shipping, is going to begin levying a carbon tax starting in 2027.

But in any place where ammonia is burned — whether that's a power plant or a oceangoing ship — it needs to have at least some fossil fuel combusted alongside it. That makes full decarbonization impossible.

Amogy has been developing a way to fully replace fossil fuels using ammonia as a fuel. First, the company cracks three hydrogen atoms off each nitrogen atom. It then sends the hydrogen to a fuel cell, which generates electricity and water vapor, while releasing pure nitrogen to the air.

Because there's no combustion, the company's process doesn't release any NO_x pollution, which can create smog and cause a host of health problems.

The startup previously tested its technology in a tug boat, and it's still on track to deploy a commercial-scale system in a ship by the next couple years. But Amogy is also developing a power plant that will provide power to terrestrial customers, including data centers. The first of its kind will begin generating power in the next couple years, Woo said.

The first systems will be on the smaller side, capable of producing 500 kilowatts to 1 megawatt of electricity, though customers can deploy several in parallel to generate more power.

Woo said that Amogy's shift to Japan and South Korea comes at a time when the countries are beginning to develop their ammonia infrastructure. By the end of the decade, coal power plants in both countries are expected to use some amount of ammonia in their operations.

Initially, the ammonia will likely come from the U.S. and the Middle East, where hydrogen bound in inexpensive natural gas is used to make the compound. Asian countries are setting standards for how much carbon pollution can result from ammonia production. As a result, it's likely that producers will need to capture at least some of the carbon to be able to sell to those markets.

But down the line, Woo said, the hope is to transition to green sources of hydrogen to create ammonia. Asian countries, Woo said, "see ammonia basically as the next LNG, but without the carbon."

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Green ammonia startup Amogy is trying to raise

$90M to reduce truck pollution

Tim De Chant 10:55 AM PST · December 11, 2024

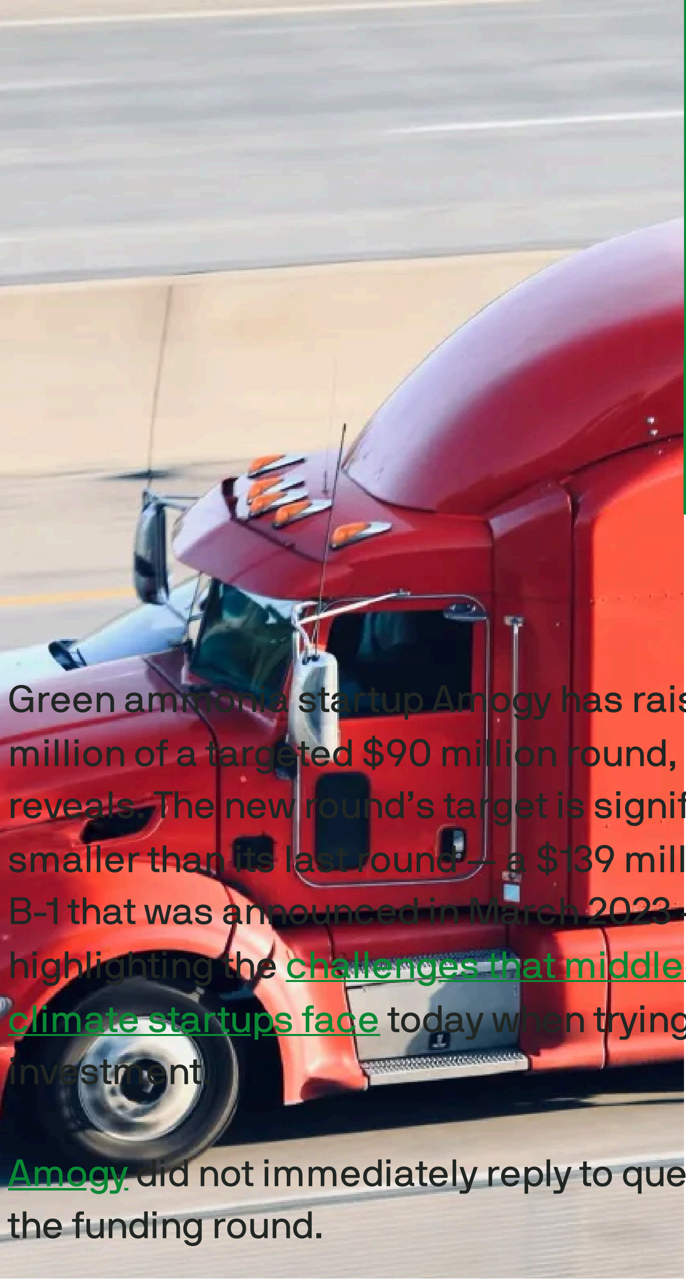

IMAGE CREDITS: DOUGLAS SACHA / GETTY IMAGES

Green ammonia startup Amogy has raised $11.2 million of a targeted $90 million round, an SEC filing reveals. The new round's target is significantly smaller than its last round — a $139 million Series B-1 that was announced in March 2023 — highlighting the challenges that middle-stage climate startups face today when trying to attract investment.

Amogy did not immediately reply to questions about the funding round.

The company is focused on various transportation markets, including maritime shipping, long-haul trucking, and agriculture. Earlier this year, Amogy tested its ammonia-to-power technology in a retrofitted tugboat, and it previously tested it in semi-tractors and farm tractors.

Many green ammonia startups are targeting long-distance transportation in part because ammonia can be burned inside traditional internal combustion engines with a few modifications, although it does require a so-called pilot fuel like diesel or biofuel to help it ignite. Depending on the pilot fuel, switching to ammonia eliminates most if not all carbon emissions, but it also generates oxides of nitrogen, a class of pollutants that contributes to smog and acid rain and can cause global warming.

Amogy takes a different approach than most. Rather than burning the ammonia, it cracks the fuel into nitrogen and hydrogen. The nitrogen is emitted as harmless nitrogen gas while the hydrogen runs through a fuel cell, which produces electricity and water. The setup doesn't require a pilot fuel, either, meaning it's zero-carbon from the start.

Despite not reusing internal combustion engines already in trucks and ships, Amogy is likely to find at least some receptive customers, particularly within maritime shipping. The industry has set a deadline of 2050 to hit net-zero carbon emissions.

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Phlair’s carbon-sucking technology could lower direct air capture’s costs

Tim De Chant — 3:00 AM PDT · September 19, 2024

IMAGE CREDITS: PHLAIR

When it comes to climate change, there’s no such thing as a “get out of jail free” card. But there might be an inexpensive alternative: direct air capture.

The technology isn’t exactly an exoneration, but more like community service; it promises to suck massive amounts of carbon dioxide out of the atmosphere, atoning for our century-plus of transgressive burning of fossil fuels. Scientifically, it’s a sound idea. Commercially, it has been less so.

Currently, it costs about $600 to $1,000 to capture a metric ton of carbon, which is far more than anyone thinks is commercially viable. So myriad startups are racing to cut costs, aiming to capture one metric ton of carbon dioxide for $100 or less.

Even at that price, it could be a difficult sell since burning fossil fuels remains, for the most part, free. But many investors and even a few multinational corporations like Microsoft, Shopify, and Stripe are

betting that eventually, the world will embrace direct air capture, much like how we treat wastewater today instead of dumping it into a river.

Larger startups like Climeworks and Carbon Engineering are betting that scale will help rein costs in. Both companies use sorbents to draw out the carbon dioxide and use heat to release it from the sorbents so it can be stored elsewhere.

Smaller startups suggest that scale alone won't be enough, though. "Thermal regeneration is always the expensive step, energy wise," said Malte Feucht, co-founder and CEO of Phlair, a young direct air capture startup. He may have a point. One study says that capturing a meaningful amount of carbon, around 10 gigatons per year, using Carbon Engineering's approach would require nearly three-quarters of all the electricity generated in the world today.

Feucht's company thinks that a different approach that doesn't rely on heat might help bring costs down. Like most direct air capture companies, Phlair uses fans to blow air over an absorber. But instead of heating the sorbent, it uses an acid to liberate the carbon dioxide. To produce the acid and base used in the process, Phlair, formerly known as Carbon Atlantis, developed a device it calls a hydrolyzer.

The hydrolyzer borrows heavily from the hydrogen industry, taking elements from both membrane-based electrolyzers and membrane-based fuel cells, Feucht said. (An electrolyzer makes hydrogen using electricity, whereas a fuel cell consumes hydrogen to produce it.)

“Instead of hydrogen, we only produce acids and bases,” he said.

Phlair’s DAC machine employs what’s known as the “pH swing” method to capture carbon dioxide. Inside, the basic (high pH) solvent absorbs carbon dioxide as it flows through the air contractor. After the saturated solvent exits the contractor, it is dumped into a tank where it’s doused with acid (low pH). That swing in pH from high to low spurs a chemical reaction that releases the carbon dioxide so it can be piped elsewhere to be used or stored. The solvent then flows back into the hydrolyzer where it’s regenerated.

Phlair is deploying a pilot in the next few weeks, Feucht said, that can capture around 10 metric tons of carbon per year. After that, the startup is working on larger, 260-metric-ton plants that are scheduled to come online in late 2025. One being built with Paebbl in the Netherlands will deliver carbon to help make a cement additive, while the other in Canada will be built with Deep Sky, a carbon removal project developer, which will store the carbon.

The DAC startup has already sold a number of carbon credits to organizations like Frontier, which works with Alphabet, Meta, Shopify, Stripe, and others to create an advanced market commitment for direct air capture.

To help complete the larger projects, Phlair has raised a €12 million seed round along with a €2.5 million grant from the EU's EIC Accelerator. Extantia Capital led the investment round with Atlantic Labs, Counteract, Planet A, UnternehmerTUM Funding for Innovators, and Verve Ventures participating.

"I think this is a sort of a unique time in history. Ten years ago, you would have probably needed to found an NGO to do what we're doing," Feucht said. "Now there's a real opportunity to serve customers, to build a functioning company, but then also to address that [carbon] problem. For me, that's my personal, super big motivation."

Topics: carbon capture and storage   Climate   direct air capture   Exantia Capital   Exclusive   Fundraising   Phlair   Startups

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